

increasingly, which are based on the flexible high-temperature plastic-like polyimide films like Kapton.

2. Copper Layer

The second layer, attached to the board, is a thin copper foil which is laminated to it through heat and adhesive. On common double-sided gadgets Copper is attached on both sides in two layers while in lower-cost electronic gadgets there may be only one copper layer on either side of the substrate. When we refer to our board as a double-sided or 2-layer board, we are referring to the number of copper layers it has which can be anywhere between one and sixteen.

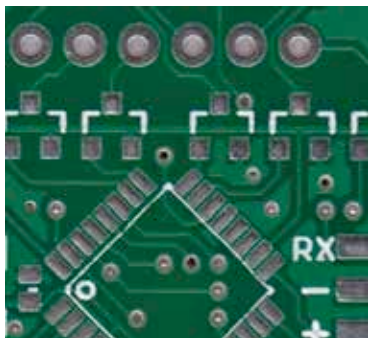


<https://cdn.sparkfun.com>

3. Soldermask

The layer that comes on top of the copper foil is called the solder mask. This layer is responsible for giving a PCB its green color, and it is overlaid on the copper layer to prevent it from contacting other metal solders or conductive bits. The primary use of this component is to help users solder on the correct places and prevent solder jumpers.

Soldermask is most commonly green in color but can be of any color at all, depending on the production.



<https://cdn.sparkfun.com>

4. Silkscreen

A white silkscreen layer is applied on top of the solder mask layer to print numbers, letters and other indicators so that it becomes easier for the user to understand the functionality of a particular pin or LED. The silkscreen